

Cyril Jacob

Software Engineer | AI/ML Systems | Linux, Kubernetes & Cloud

+91 88741 88732 | itscyriljacob@gmail.com | cj445.github.io | linkedin.com/in/thecyriljacob | github.com/cj445 | New Delhi, India

Summary

Computer Science (AI & ML) student graduating in 2027 who builds and deploys AI systems end to end, from data preparation and deep learning model development to optimization with deep learning compilers (TensorRT, ONNX) and containerized deployment on Linux. One year of industry experience with a US-based company across RAG data pipelines, cloud infrastructure, and DevOps.

Skills

AI/ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, Computer Vision (OpenCV, YOLOv8, Mask R-CNN, CLIP)

Deep Learning Compilers & Optimization: TensorRT, ONNX, INT8 Quantization, NVIDIA DeepStream SDK, Edge Deployment

NLP & Retrieval: Retrieval-Augmented Generation (RAG), Vector Search (ChromaDB, HNSW Indexing)

Programming: Python, SQL, Shell Scripting

Databases & Data: PostgreSQL, MongoDB, Redis, MySQL, Apache Kafka, MQTT

Hardware: Arduino Uno, Raspberry Pi, NVIDIA Jetson Orin Nano, Eagle 101 Nano, ZED X Stereo Camera

Linux & Cloud: Linux (Ubuntu), Kubernetes, Docker, Azure, Azure DevOps (CI/CD), REST APIs, Microservices, Git, GitHub

Work Experience

Software Engineering Intern

Jan 2025 – Dec 2025

Graceful Management Systems (US-based, Remote)

- Designed database schemas and data preparation pipelines for a RAG application, integrating, cleansing, and shaping source data for downstream retrieval and inference workflows.
- Deployed containerized microservices to Azure; built CI/CD workflows in Azure DevOps with automated testing, self-hosted Ubuntu agents, and Azure Container Registry.
- Automated Ubuntu server provisioning with Netplan and shell scripting, reducing deployment time by 70%.
- Administered Linux servers and virtual machines, including permissions, secure SSH access, and dual-interface static IP routing; collaborated remotely with a US-based team across time zones.

Computer Vision Engineer (Trainee)

Jul 2024 – Dec 2024

Karunya Innovation and Design Studio

- Optimized YOLOv8 and Mask R-CNN models with the TensorRT deep learning compiler using INT8 quantization, reducing inference latency by 60%.
- Deployed real-time CCTV analytics on NVIDIA DeepStream SDK, improving streaming throughput by 40% (15 → 21 FPS).
- Managed Git/GitHub collaboration workflows across a multi-member engineering team.

Projects

Optical-Guided Thermal Super-Resolution

Jan 2025 – Present

Smart India Hackathon 2025 Grand Finale (Top 5), ISRO Track

- Built a physics-guided, self-supervised deep learning framework for $2\times-4\times$ thermal super-resolution using gated optical-thermal fusion and a radiance-domain physics constraint; evaluated with SSIM, PSNR, and RMSE.
- Quantized and deployed the model on NVIDIA Jetson for edge inference.

IoT Fleet Management Platform

Oct 2025 – Nov 2025

- Built a distributed edge orchestration platform for 100+ Raspberry Pi nodes using containerized microservices, REST APIs, and MQTT, backed by PostgreSQL, MongoDB, and Redis.
- Implemented OTA updates with rollback, real-time telemetry, remote commands, and device health monitoring.

Real-Time Occupancy Analytics System

Sep 2024 – Dec 2024

- Streamed live CCTV detection data through Kafka into PostgreSQL and delivered floor-wise occupancy insights via Grafana dashboards, all containerized with Docker.

Achievements & Leadership

- 4th Place Nationally, Bharatiya Antariksh Hackathon 2025 (ISRO):** Team Lead among 61,000+ students across 8,744 teams; achieved 41.45 dB PSNR and 0.97 SSIM on the official ISRO test set.
- Campus Lead, Google Developer Groups On Campus, Karunya (2024–2025):** Led a 25-member community; organized workshops, developer events, and a state-level hackathon.

Education

Karunya Institute of Technology & Sciences, Coimbatore | B.Tech, Computer Science Engineering (AI & ML) | CGPA

7.75/10 | 2023 – 2027

Publications & Certifications

Publications: *Physics-Guided Residual Super-Resolution for Thermal Infrared Satellite Imagery* — accepted at IEEE ICECA 2026; *Smart Security Management using IoT and HC-05 Bluetooth Module* — IEEE, 2024

Certifications: Microsoft Certified: Azure Fundamentals; Duke University: RAG; NVIDIA: Deep Learning and AI on Jetson Nano; SnowPro Associate; University of London: Machine Learning for All